

SHIVAM KUMAR YADAV

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist specializing in **Generative AI, Agentic Systems, and Production ML Pipelines**. Proven expertise in building end-to-end AI solutions—from RAG-powered intelligent agents to real-time deep learning applications—with measurable business impact. Skilled at translating complex ML problems into scalable, deployment-ready systems using modern LLM frameworks, computer vision, and MLOps best practices. Delivered 10+ production-grade projects achieving 85-92% model accuracy and sub-second inference times.

TECHNICAL EXPERTISE

AI & Machine Learning Frameworks

PyTorch · TensorFlow · Scikit-Learn · XGBoost · LightGBM · Keras · Hugging Face Transformers

Generative AI & LLM Stack

LangChain · RAG Systems · LLaMA-3 · OpenAI GPT · Fine-Tuning · Prompt Engineering · OpenRouter

Agentic AI & Orchestration

CrewAI · n8n · Agentic Workflows · Multi-Agent Systems · Tool-Use Integration

Deep Learning Architectures

CNN · RNN · LSTM · Transfer Learning · YOLO · Computer Vision

NLP & Text Processing

SpaCy · NLTK · TF-IDF · Transformers · Sentiment Analysis · Text Classification

Data Engineering & Analytics

Python · SQL · MySQL · Pandas · NumPy · SciPy · Statistical Modeling · Time Series Analysis

MLOps & Deployment

MLflow · GitHub · GitHub Copilot · Flask · Streamlit · Model Versioning · CI/CD Pipelines

Visualization & BI Tools

Power BI · Tableau · Seaborn · Matplotlib

Cloud & Development

MS Azure · Google Colab · VS Code · PyCharm · Jupyter Notebook · Spyder

PROFESSIONAL EXPERIENCE

AI & Machine Learning Engineer | Intensive Data Science Program

Advanced AI Systems & Production ML Development

Generative AI & Agentic Systems

- Architected RAG-powered systems using LangChain and LLaMA-3, processing 1,000+ documents with 40% faster query resolution.
- Designed multi-agent workflows with CrewAI and n8n, automating decision pipelines and reducing manual intervention by 60%.
- Fine-tuned LLMs for domain-specific tasks, improving response relevance by 25% through prompt engineering.

Production ML & Deep Learning

- Built end-to-end ML pipelines processing 10,000+ records, achieving 85-92% accuracy in forecasting and sentiment analysis.
- Developed real-time computer vision systems using PyTorch and YOLO with 90%+ accuracy at 20-25 FPS (6-8 object detection).
- Implemented NLP solutions with Transformers and SpaCy, classifying 1,000+ documents with 40% reduced preprocessing time.

MLOps & Deployment

- Established MLOps infrastructure with MLflow and GitHub Actions CI/CD, reducing deployment cycles by 70%.
- Applied hyperparameter optimization improving model performance by 18% while maintaining efficiency.

Data Engineering & Analytics

- Built ETL pipelines with Python and SQL, implementing hypothesis testing across 8+ business-critical ML projects.
- Leveraged time series analysis to validate predictive models with 95% confidence intervals.

- Created interactive dashboards in Power BI and Tableau, translating ML metrics into actionable business.

LEADERSHIP EXPERIENCE

Commanding Officer, NSS Unit

2024 – 2025

Chouksey Engineering College

- Successfully led 100+ volunteers, managing various tasks and responsibilities for NSS activities across multiple community initiatives.
- Coordinated 4 one-day camps and one 7-day camp, ensuring successful execution and stakeholder engagement.
- Enhanced teamwork, communication, and problem-solving skills by leading large groups and organizing impactful community service events.

SIGNATURE PROJECTS

Enterprise Contract Intelligence Platform | LegalTech

- Engineered AI-powered contract analysis system reducing legal document review from 4+ hours to under 2 minutes for 200-page M&A agreements with clause-specific retrieval and page-level citations.
- Architected production RAG pipeline with semantic chunking, vector embeddings, and context-aware retrieval serving law firms and paralegal teams with 95%+ extraction accuracy.
- Solved real-world PDF parsing challenges (scanned documents, multi-column layouts, embedded tables) through intelligent fallback architecture ensuring reliable processing of messy legal documents.
- Implemented hallucination prevention via temperature tuning and source verification, delivering legally defensible AI-generated insights for time-sensitive due diligence.

Business Impact: Enables legal teams to perform contract due diligence 120x faster—transforming review bottlenecks into competitive advantage.

Strategic Intelligence Automation System | Market Research

- Built autonomous research platform replacing 8-12 hour manual market analysis with AI agent orchestration, delivering comprehensive competitive intelligence reports in under 5 minutes.
- Designed multi-agent architecture where Planner decomposes complex queries, Search agents gather data from 15-20 sources in parallel, and Synthesizer produces board-ready reports with citations.
- Engineered enterprise-grade reliability with graceful API degradation ensuring 100% system availability when external data sources fail—critical for time-sensitive strategic decisions.
- Transformed raw web data (competitor sites, market reports, news) into actionable strategic recommendations with confidence scoring.

Business Impact: Empowers C-suite executives and strategy consultants to make data-driven decisions 96x faster—replacing weeks of analyst work with on-demand intelligence.

Medical Accessibility AI Platform | HealthTech

- Developed medical document intelligence system enabling instant translation and OCR extraction of prescriptions, pharmacy labels, and medical notes for 200M+ non-English speakers globally.
- Engineered production vision pipeline processing handwritten prescriptions, low-resolution labels, and multi-script documents (Telugu, Tamil, Arabic, Chinese) with 88%+ accuracy under adverse conditions.
- Implemented healthcare-compliant AI architecture with safety disclaimers, confidence scoring, and fail-safe mechanisms—balancing innovation with patient safety in high-stakes medical contexts.
- Built edge-case resilient system handling 50MB+ scans, motion-blurred photos, and mixed-language documents common in multicultural healthcare settings.

Business Impact: Prevents medication errors for immigrant populations—addressing WHO-recognized language barriers causing 60% of preventable adverse drug events.

EDUCATION

Bachelor of Technology (B.Tech)

2025

Chouksey Engineering College

XII (CBSE)

2021

Burgess English Higher Secondary School

CERTIFICATIONS & ACHIEVEMENTS

Data Science Professional – Oracle

Data Analytics Job Simulation – Deloitte

Machine Learning – IBM

AIML Data Analysis Workshop – ISRO (7-Day)

Generative AI Professional – Oracle

AI Skills Passport – EY & Microsoft

NSS 'C' Certificate – CSVTU (National Level)